

Week 06: Correlation Analysis

Introduction:

This analysis explores the correlation between 2 numerical variables in the gss data set, namely, hrsonline and hrsworked

	id	hrsworked	childs	age	educ	hrsonline	liberal	overtime	married
count	98.000000	98.000000	98.000000	98.000000	98.000000	98.000000	98.000000	98.000000	98.000000
mean	1361.653061	41.622449	1.357143	40.204082	14.826531	5.765306	0.275510	0.387755	0.530612
std	805.223669	14.795872	1.348921	11.768348	2.773310	7.177860	0.449068	0.489743	0.501628
min	43.000000	2.000000	0.000000	20.000000	8.000000	0.000000	0.000000	0.000000	0.000000
25%	616.000000	37.750000	0.000000	30.000000	12.000000	1.000000	0.000000	0.000000	0.000000
50%	1344.000000	40.000000	1.000000	41.000000	14.000000	3.000000	0.000000	0.000000	1.000000
75%	2078.750000	49.500000	2.000000	49.000000	17.000000	7.000000	1.000000	1.000000	1.000000
max	2700.000000	89.000000	7.000000	69.000000	20.000000	35.000000	1.000000	1.000000	1.000000

Correlation Coefficients

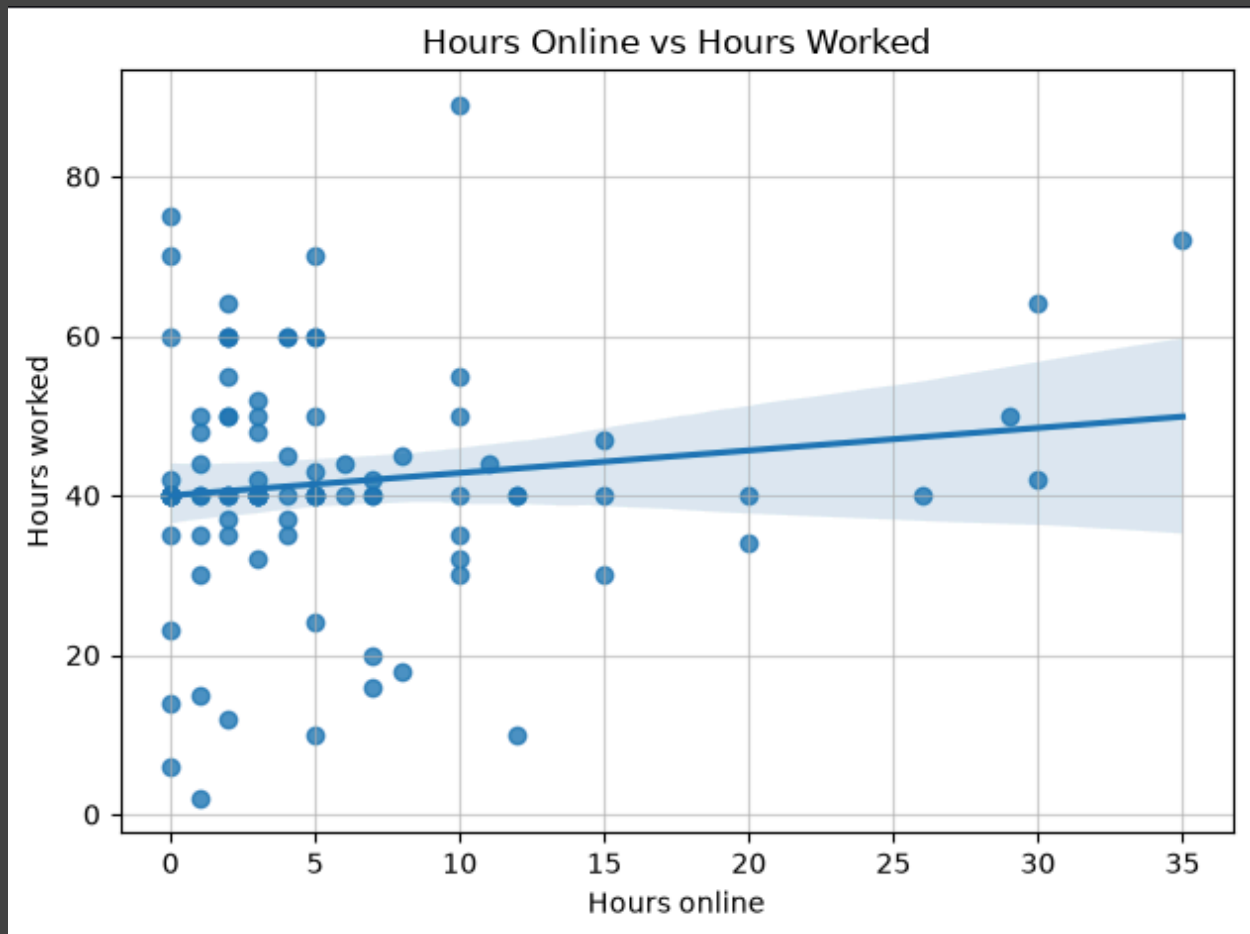
Method	Coefficient	p-value	Statistically significant at $\alpha = 0.05$?
Pearson	0.137	0.179	False
Spearman	0.055	0.591	False
Kendall	0.041	0.588	False

The results indicate a weak positive relationship between hours online and hours worked. The values of the correlation coefficients are small though and none of the correlations is statistically significant since all p-values are greater than 0.05.

Coefficient of determination $R^2 = 0.0188$ or 1.88% suggests that the linear relationship represented by Pearson's correlation accounts for 1.88% only of the variations in hours worked

Normality assumption concerns

There could be some concerns about the normality assumption underlying the Pearson significance test. Hours online appear to be right-skewed for the mean = 5.765 > the median at 3.00, and a max of 35 hours, suggesting some higher values pulling the mean to the right of the median. However, non parametric correlations checked, Spearman and Kendall support the conclusion that there is no significant evidence of association between hours online and hours worked in the studied sample.



The trend line has a slight upward slope which matches is consistent with the positive pearson correlations $r = 0.137$.

The regression equation is $y = 0.282x + 39.994$, and the slope of 0.282 indicates that an additional hour spent online is associated with 0.282 additional predicted hours worked.

However the values are widely dispersed around the line, consistent with the weak Pearson correlation, also with some values of hours online on the far right end of the plot supporting the right-skewness of the hours online variable as well.

Point bi-serial correlation

Method	Coefficient	p-value	Statistically significant at $\alpha = 0.05$?
Point bi-serial	-0.185	0.068	False

married	count	mean
0	46	15.37
1	52	14.35

There is a negative point biserial correlation between marital status and years of education $r_{pb} = -0.185$. Participants who are married had lower mean years of education at 14.35, while unmarried participants recorded higher mean at 15.37.

However, $p = 0.068 > 0.05$, suggesting the correlation is not statistically significant, i.e. not providing enough evidence against the null hypothesis ($H_0: \rho=0$) of no correlation between years of education and the marital status.

Controlling for age

After fitting the data with a linear curve, predicted and residual values were calculated then pearson correlation was applied to the residuals (i.e the values that are not explained by age)

	hrsonline	hrsworked	age	hrsonline_predicted	hrsworked_predicted	hrsonline_residual	hrsworked_residual
0	3	40	42	5.742405	41.398867	-2.742405	-1.398867
1	5	10	52	5.614887	40.153922	-0.614887	-30.153922
2	5	43	52	5.614887	40.153922	-0.614887	2.846078
3	5	70	23	5.984688	43.764263	-0.984688	26.235737
4	1	2	59	5.525625	39.282460	-4.525625	-37.282460

Partial correlation: $r = 0.136$, $p = 0.1830$

	r_pearson	p
Original value	0.137	0.1786
After controlling for age	0.136	0.1830

Controlling for age has almost no effect on the relationship. The difference in the correlation coefficient is negligible at 0.001 which suggests the age had very little impact on the relationship between hours online and hours worked, and the partial correlation is still statistically insignificant with p at $0.183 > 0.05$.